

DISCOVERING HIDDEN RELATIONS AMONG TRIANGLE INVARIANTS

TRIANGLE-RELATIONS PROJECT

ABSTRACT. We describe a numerical method for discovering unknown functional relations among scalar quantities derived from a triangle, in the spirit of Euler’s relation between the circumradius, the inradius, and the incenter-to-circumcenter distance. A triangle has three degrees of freedom, so a relation among four or more derived scalars is guaranteed by dimension counting alone and is uninformative; a relation among exactly three is a genuine, generically unexpected coincidence. We detect such coincidences by training a bottleneck autoencoder on sampled triangle data and comparing its reconstruction error against a column-permuted null, and make the detector’s behavior mathematically precise. Applied to (R, r, d) , the method flags exactly this triple as dependent, with no prior knowledge of Euler’s relation.

1. DIMENSION COUNTING

A triangle, up to translation and rotation, has three degrees of freedom (e.g. its three side lengths a, b, c); write $\mathcal{M}$ for this moduli space, $\dim \mathcal{M} = 3$. Every derived scalar — area, the circumradius R (the radius of the circle through all three vertices), the inradius r (the radius of the circle inscribed in the triangle), distances between derived points, ... — is a smooth function $f : \mathcal{M} \rightarrow \mathbb{R}$, and k of them give a map $F = (f_1, \dots, f_k) : \mathcal{M} \rightarrow \mathbb{R}^k$. Call a property of F *generic* if it holds outside a closed, measure-zero exceptional set of choices of the f_i ; for instance, the Jacobian $DF(p)$ (the $k \times 3$ matrix of partial derivatives $(\partial f_i / \partial x_j)$ in local coordinates on $\mathcal{M}$ at a point p) generically has the largest rank its shape allows, $\min(k, 3)$, since failing to do so is a closed, codimension- ≥ 1 condition on the f_i .

If $k > 3$, take a point p where $DF(p)$ has the generic rank 3, i.e. F is an immersion at p : by the constant rank theorem, in suitable local coordinates near p and $F(p)$, F looks like the standard inclusion $\mathbb{R}^3 \hookrightarrow \mathbb{R}^k$. Consequently the image of a small neighborhood of p is a smooth 3-dimensional submanifold, cut out there by $k - 3$ independent equations $G_i(f_1, \dots, f_k) = 0$ in the $k - 3$ coordinate directions transverse to it. These $k - 3$ local relations arise from dimension counting alone, *for free*, regardless of which f_i were chosen; a relation among four or more triangle invariants therefore says nothing specific about triangle geometry.

If $k = 3$, recall F is a *local diffeomorphism* at p (a smooth bijection near p with smooth inverse) exactly when $DF(p)$ is invertible, by the inverse function theorem. Generically DF has full rank 3, and near such a point (f_1, f_2, f_3) sweeps out an

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open subset of $\mathbb{R}^3$ with nonempty interior — a genuine 3-dimensional volume, not a lower-dimensional surface. An *exact* relation $G(f_1, f_2, f_3) = 0$ holding identically is equivalent to DF having rank ≤ 2 everywhere, collapsing the image onto a 2-dimensional surface: not generic, a genuine algebraic coincidence. Euler's relation

$$d^2 = R^2 - 2Rr \quad (1.1)$$

(where d is the incenter-to-circumcenter distance) is exactly such a statement about (R, r, d) . This is the sense in which a relation among exactly three invariants — not four — is structurally new.

Is the solution set of Euler's relation actually a submanifold, or does it have edges or cusps? Generically the former, with one exception.

Proposition 1.1 (Smoothness of the Euler surface). *Let $\Sigma = \{(R, r, d) \in \mathbb{R}^3 : d^2 = R^2 - 2Rr\}$. Then Σ is a smooth (real-analytic) embedded 2-dimensional submanifold of $\mathbb{R}^3$ away from the single point $0 = (0, 0, 0)$, which is a genuine conical singularity: writing $u = R - r$, Euler's relation becomes $u^2 = r^2 + d^2$, the equation of a standard double cone with vertex at $u = r = d = 0$. No actual (nondegenerate) triangle reaches this vertex, since every triangle has $R > 0$ strictly, so the physically realized part of Σ is smooth throughout. It does have one edge — not a cusp — along the equilateral locus $r = R/2$, $d = 0$, where Euler's inequality $R \geq 2r$ becomes an equality.*

Proof. The gradient of $G(R, r, d) = R^2 - 2Rr - d^2$ is $\nabla G = (2R - 2r, -2R, -2d)$, which vanishes only when $R = r$, $R = 0$, $d = 0$ simultaneously, i.e. only at the origin (which does lie on Σ , since $G(0, 0, 0) = 0$). By the implicit function theorem, Σ is a smooth submanifold at every point where $\nabla G \neq 0$, hence everywhere except possibly the origin. Completing the square, $G = (R - r)^2 - r^2 - d^2$, so with $u = R - r$, $\Sigma = \{u^2 = r^2 + d^2\}$: the standard double cone. A punctured neighborhood of its vertex $u = r = d = 0$ on Σ has two connected components (one per nappe, since $u = 0$ forces $r = d = 0$ too), unlike a punctured disk in $\mathbb{R}^2$; so the vertex is not even a topological manifold point, confirming it is a genuine singularity.

For an actual triangle, $R > 0$ always (a nondegenerate triangle has positive circumradius), so the physically realized points of Σ never coincide with the singular vertex. Writing $d = \sqrt{R(R - 2r)}$ (the nonnegative root, since d is a distance) over the physical domain $\{R > 0, 0 < r \leq R/2\}$ — using Euler's inequality $R \geq 2r$, which always holds for a genuine triangle — exhibits the physical surface as a smooth graph, with an edge exactly at $r = R/2$ ($d = 0$, the equilateral case). Viewed on the full cone Σ , this locus is an ordinary smooth curve: $\nabla G \neq 0$ there too, e.g. at $(R, R/2, 0)$ with $R > 0$, $\nabla G = (R, -2R, 0) \neq 0$. It becomes an edge of the *physical* surface only because the mirror branch $d < 0$ — equally part of Σ , but physically meaningless — is discarded. $\square$

Figure 1 shows a representative local patch of the physical surface together with sampled triangles, which fall exactly on it, with the equilateral edge marked in red; Figure 2 shows the full variety Σ , both branches, meeting at the singular vertex. (Since Σ is a cone, plotting it from R near 0 up to some $R_{\max}$ in a single view would make it look like a thin flared blade rather than a curved sheet, an artifact of showing many scales at once rather than a feature of the surface itself; Figure 1 instead zooms into one scale.)

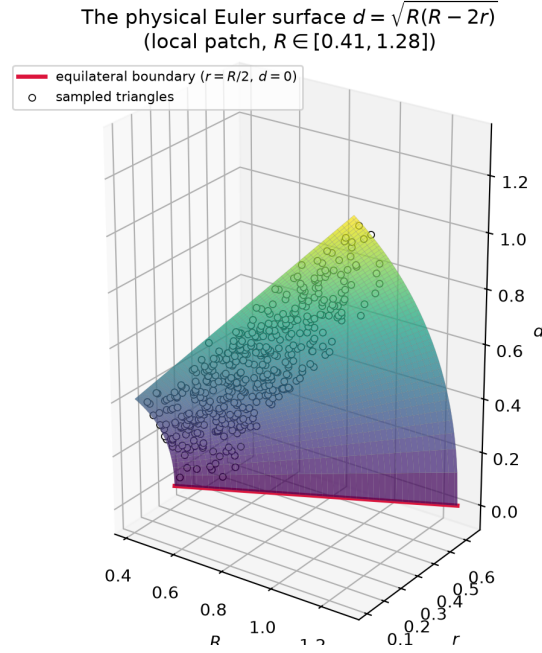


FIGURE 1. The physical Euler surface $d = \sqrt{R(R - 2r)}$, shown over a representative local patch of R -values, with sampled triangles (white dots) lying exactly on it. The red line is the equilateral edge $r = R/2$, $d = 0$.

2. THE AUTOENCODER DETECTOR

Given a candidate triple (f_1, f_2, f_3) , sample many random triangles and form the point cloud $X = \{(f_1, f_2, f_3)(T)\} \subset \mathbb{R}^3$. An autoencoder — an encoder $e : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ and decoder $d : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ trained to minimize $\mathbb{E} \|d(e(x)) - x\|^2$ — searches for the best rank-2 approximation to X : if X lies on a 2-surface, the encoder need only learn a local chart and the decoder its inverse, driving error to 0; if X fills an open 3-region, no map factoring through $\mathbb{R}^2$ can avoid discarding a whole dimension, so error stays bounded away from 0.

2.1. A precise statement. Let P be the law of $(f_1, f_2, f_3)(T)$ and $X = \text{supp}(P)$. Fix input and output dimensions $p, q \geq 1$ (for us, always $p, q \in \{2, 3\}$). By a *two-layer perceptron of width h* we mean a map $N : \mathbb{R}^p \rightarrow \mathbb{R}^q$ of the form $N(x) = W_2 \sigma(W_1 x + b_1) + b_2$, with $W_1 \in \mathbb{R}^{h \times p}$, $b_1 \in \mathbb{R}^h$, $W_2 \in \mathbb{R}^{q \times h}$, $b_2 \in \mathbb{R}^q$ trainable and $\sigma = \tanh$ applied componentwise; write $\mathcal{N}_h(\mathbb{R}^p, \mathbb{R}^q)$ for the class of all such maps $\mathbb{R}^p \rightarrow \mathbb{R}^q$. Recall g is K -Lipschitz if $\|g(x) - g(y)\| \leq K\|x - y\|$; since $\tanh$ is 1-Lipschitz, every two-layer perceptron is Lipschitz, with constant controlled by $\|W_1\|, \|W_2\|$. Train an encoder $e \in \mathcal{N}_h(\mathbb{R}^3, \mathbb{R}^2)$ and a decoder $d \in \mathcal{N}_h(\mathbb{R}^2, \mathbb{R}^3)$ to minimize $L(e, d) = \mathbb{E}_{x \sim P} \|d(e(x)) - x\|^2$.

Proposition 2.1 (Sufficiency). *Recall a subset of $\mathbb{R}^n$ is compact iff closed and bounded (Heine–Borel). If X is compact and admits continuous $\varphi : X \rightarrow \mathbb{R}^2$, $\psi : \varphi(X) \rightarrow \mathbb{R}^3$ with $\psi(\varphi(x)) = x$ for all $x \in X$ — in particular whenever an exact relation confines X to a 2-dimensional set — then for every $\varepsilon > 0$ there is a width h and (e, d) with $L(e, d) < \varepsilon$.*

Both branches of $d^2 = R^2 - 2Rr$
 physical: green ($R > 0$) unphysical: gray ($R \leq 0$)

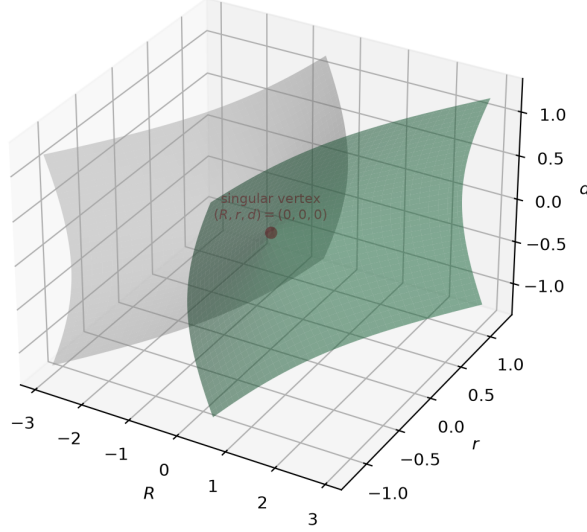


FIGURE 2. Both algebraic branches of $d^2 = R^2 - 2Rr$: the physical branch $R = r + \sqrt{r^2 + d^2}$ (green) and the unphysical branch $R = r - \sqrt{r^2 + d^2} \leq 0$ (gray), meeting only at the singular vertex $(R, r, d) = (0, 0, 0)$.

Proof sketch. By the universal approximation theorem (Cybenko 1989; Hornik 1991): for any continuous $\varphi : K \rightarrow \mathbb{R}^q$ on compact K and any $\delta > 0$, some $N \in \mathcal{N}_h(\mathbb{R}^p, \mathbb{R}^q)$ satisfies $\sup_{x \in K} \|N(x) - \varphi(x)\| < \delta$. Approximating φ on X and ψ on $\varphi(X)$ gives e, d with $\sup_{x \in X} \|d(e(x)) - x\| \rightarrow 0$, hence $L(e, d) \rightarrow 0$. $\square$

In plain terms: if (f_1, f_2, f_3) really do satisfy an exact relation, a large enough autoencoder reconstructs them essentially perfectly, however the relation happens to curve through $\mathbb{R}^3$ — the encoder just learns to read off a valid pair of local coordinates on that curved surface, and the decoder learns to invert them. The next proposition is the converse: without such a relation, no autoencoder of bounded complexity can do better than some fixed, positive error, no matter how it is trained. “Bounded complexity” here means a bound on the Lipschitz constant K of e and d (roughly, how fast their outputs can change relative to their inputs); we explain right after the proof why this qualifier is unavoidable.

Proposition 2.2 (Necessity, at bounded capacity). *Fix $K > 0$. Recall P is absolutely continuous (w.r.t. Lebesgue measure) if it assigns zero probability to every Lebesgue-null set, i.e. has a density. If X contains an open ball B on which P is absolutely continuous with $P(B) > 0$, there is $c = c(P|_B, K) > 0$ with $L(e, d) \geq c$ for every pair of K -Lipschitz e, d — in particular for every $(e, d) \in \mathcal{N}_h(\mathbb{R}^3, \mathbb{R}^2) \times \mathcal{N}_h(\mathbb{R}^2, \mathbb{R}^3)$ whose realized weights keep the Lipschitz constant below K .*

Proof sketch. The Hausdorff dimension of $S \subset \mathbb{R}^n$ generalizes ordinary dimension to non-smooth sets (covering S by ε -balls takes $\sim \varepsilon^{-\dim S}$ of them); Lipschitz maps do not increase it. So $d(\mathbb{R}^2)$ has Hausdorff dimension ≤ 2 , hence is Lebesgue-null

in $\mathbb{R}^3$, hence P -null on B ; thus $\text{dist}(x, d(\mathbb{R}^2)) > 0$ for $P|_B$ -a.e. x , and a compactness argument over K -Lipschitz maps gives $\mathbb{E}_{x \sim P|_B} \text{dist}(x, d(\mathbb{R}^2))^2 \geq c > 0$. Since $d(e(\mathbb{R}^3)) \subseteq d(\mathbb{R}^2)$, the same bound applies to $L(e, d)$. $\square$

Remark 2.3. By Brouwer’s invariance-of-domain theorem, a continuous injection between open subsets of $\mathbb{R}^n$ is an open map, so no continuous injection of $\mathbb{R}^2$ into $\mathbb{R}^3$ has open image: exact space-filling is topologically impossible regardless of capacity. What is not ruled out is *approximate* space-filling by increasingly large-Lipschitz-constant maps as $K \rightarrow \infty$, which is why Proposition 2.2 fixes K ; in practice ℓ_2 weight regularization and early stopping keep the realized Lipschitz constant controlled.

Propositions 2.1 and 2.2 say that, at any fixed, regularized capacity, reconstruction error going to 0 versus staying bounded away from it is decided entirely by whether (f_1, f_2, f_3) satisfy an exact relation — the theoretical content behind the detection rule used throughout. Figure 3 shows this dichotomy on the actual triple (R, r, d) .

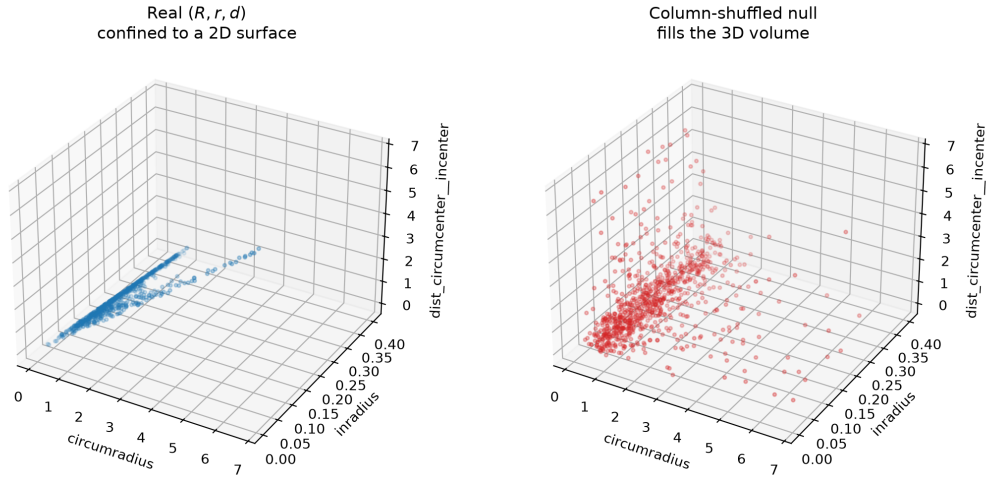


FIGURE 3. The real triple (R, r, d) (left, 1000 sampled triangles) collapses onto a thin 2-dimensional surface, as Proposition 2.1 predicts; independently shuffling each column (right) destroys the relation, and the same points fill an open 3-dimensional region, as Proposition 2.2 predicts.

3. CALIBRATION VIA A PERMUTATION NULL

Propositions 2.1 and 2.2 describe an idealized limit ($h \rightarrow \infty$, exact expectations); in practice we train one finite network on one finite sample, and its reconstruction loss L_{real} is never exactly 0 or exactly the theoretical floor c — it is a single noisy number. An absolute threshold on L_{real} (“flag it as dependent if the loss is below 0.01”, say) is therefore not meaningful: different triples of scalars differ in units and scale, network training is itself noisy, and, since all derived scalars are functions of the same three underlying triangle parameters, even functionally *independent* triples can be mildly correlated and so reconstruct a bit better than one might naively expect. What we

need is not one global cutoff but a baseline for “no relation” calibrated separately for each triple.

We obtain that baseline by destroying exactly the joint structure between the three scalars while leaving everything else about the data unchanged. Concretely, write the sample as a matrix $X \in \mathbb{R}^{n \times 3}$, one row per sampled triangle. Form a *shuffled* copy X' by independently and uniformly permuting the n entries within each of the three columns (a different random permutation per column). Each column of X' is some reordering of the corresponding column of X , so it has exactly the same values, and hence the same one-dimensional marginal distribution, as before; but the pairing *across* columns is now uniformly random, so whatever functional relationship held between f_1, f_2, f_3 in the real data has been destroyed. Training the same encoder/decoder architecture on X' therefore measures how well an autoencoder can compress three quantities that merely share these marginal distributions but have no functional relationship at all — precisely the situation of Proposition 2.2.

We repeat the shuffle-and-train step m times (an independent random reshuffling of X each time), obtaining m null reconstruction losses $L'_1, \dots, L'_m$, and record their sample mean and sample standard deviation,

$$\mu_{\text{null}} = \frac{1}{m} \sum_{i=1}^m L'_i, \quad \sigma_{\text{null}} = \left(\frac{1}{m-1} \sum_{i=1}^m (L'_i - \mu_{\text{null}})^2 \right)^{1/2}. \quad (3.1)$$

These summarize the null distribution of reconstruction loss for this particular triple’s marginals: μ_{null} is its typical value under “no relation”, and σ_{null} is how much that value fluctuates from one reshuffling (and training run) to the next. We compare the real loss to this null distribution via the standard z-score,

$$z = \frac{\mu_{\text{null}} - L_{\text{real}}}{\sigma_{\text{null}}}, \quad (3.2)$$

i.e. the number of null standard deviations by which the real loss falls below the null mean, or equivalently via the ratio $L_{\text{real}}/\mu_{\text{null}}$. A large z (ratio close to 0) means the real triple reconstructs far better than shuffled versions of itself typically do: the fingerprint of Proposition 2.1 taking over, and our evidence of an unexpected functional relation. A z near 0 (ratio close to 1) means the real data compresses no better than its own null, consistent with Proposition 2.2 and no relation. Because μ_{null} and σ_{null} are computed separately for each candidate triple, from data with the same units, scale, and marginal shape as the real sample, this comparison is self-calibrating and needs no triple-independent threshold.

Figure 4 shows this construction on (R, r, d) itself, with $m = 20$ reshufflings: L_{real} sits well to the left of every one of the m null losses, and in particular below the $\mu_{\text{null}} - \sigma_{\text{null}}$ edge of the shaded band.

4. THE PIPELINE

- (1) Enumerate candidate triples of derived scalars.
- (2) Sample random triangles once per triple; train a 2-bottleneck autoencoder; compare held-out error to a column-shuffled null (several repeats).
- (3) Rank triples by real/null ratio (small = strong candidate).

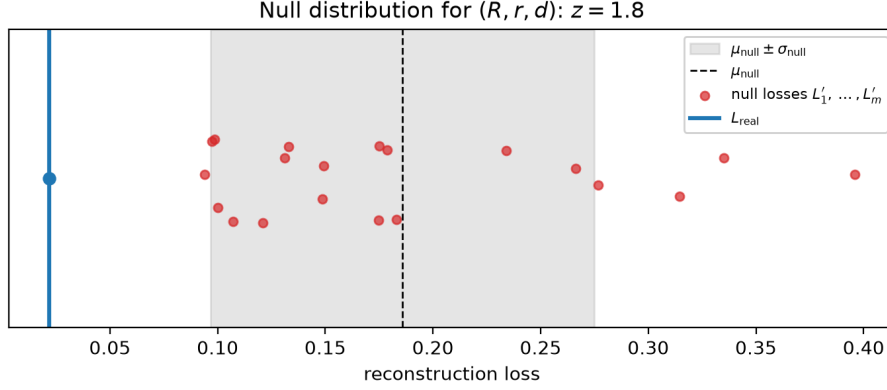


FIGURE 4. The null distribution of reconstruction loss for (R, r, d) : $m = 20$ column-shuffled losses $L'_1, \dots, L'_m$ (red), their mean μ_{null} and ± 1 standard-deviation band (dashed line, shaded), and the real loss L_{real} (blue), which lands clearly below the entire null cluster.

The autoencoder is purely a screening tool: it flags *whether* a triple is suspiciously dependent, not what the relation is. The `verify_euler_relation` module runs this on (R, r, d) , confirming it flags Euler’s relation with no prior knowledge of the formula.

5. A SCALE-FREE REFORMULATION VIA SHAPE SPACE (PROGRAM 1B)

The permutation null of Section 3 exists to answer one question: how small must L_{real} be, in the arbitrary units and scale of a given triple (f_1, f_2, f_3) , to count as evidence of a relation? Increasing the number of reshufflings m improves that per-triple calibration but is exactly the expensive step: each reshuffling retrains an autoencoder from scratch. This section describes an alternative, restricted to *homogeneous* relations, in which the reconstruction error is intrinsically bounded — the same bound for every triple — so no calibration step is needed at all. It is implemented as “Program 1b” in `triangle_relations.discovery.homogeneous_relations` and `triangle_relations.discovery.shape_space`, alongside (not instead of) Program 1 from the preceding sections.

5.1. The shape sphere. The group $\mathbb{R}_{>0}$ acts on $\mathcal{M}$ by uniform dilation (scaling a representative triangle about its centroid by $\lambda > 0$); this is well-defined on $\mathcal{M}$ since dilation commutes with the translation and rotation already quotiented out. Write

$$\Sigma = \mathcal{M} / \mathbb{R}_{>0} \quad (5.1)$$

for the quotient, $\dim \Sigma = 2$: this is the space of triangle *shapes*, forgetting size entirely. For triangles with labeled vertices, a classical theorem of Kendall [1] identifies Σ diffeomorphically with a round 2-sphere (of radius $\frac{1}{2}$ in the standard normalization), the two poles corresponding to the positively- and negatively-oriented equilateral triangle and the equator to the degenerate (collinear) triangles. In particular Σ is compact, with a canonical, bounded geodesic distance.

Remark 5.1 (An explicit chart on shape space). Kendall's theorem identifies Σ with S^2 abstractly; to actually evaluate f_1, f_2, f_3 we need a concrete triangle, so an explicit chart is unavoidable — no code can accept "a point of an abstract 2-manifold" as input. Fix two of the three labeled vertices as a base and normalize by the unique orientation-preserving similarity sending them to $0, 1 \in \mathbb{C} \cong \mathbb{R}^2$: the third vertex lands at some $\zeta = (C - A)/(B - A) \in \mathbb{C}$, and conversely every ζ gives back the concrete triangle $(0, 0), (1, 0), (\operatorname{Re} \zeta, \operatorname{Im} \zeta)$. This is exactly a chart on Σ (identifying it with the Riemann sphere $\mathbb{CP}^1$ via ζ), covering every shape except the single degenerate configuration $B = A$ (i.e. $\zeta = \infty$). Sampling ζ uniformly on $\mathbb{C}$ would not sample Σ evenly, since the chart stretches near infinity; instead `triangle_relations.discovery.shape_space` places points evenly on S^2 (a Fibonacci lattice, a standard low-discrepancy construction) and pushes them through the inverse of the stereographic projection defining ζ , giving deterministic, even coverage of Σ rather than relying on i.i.d. sampling to eventually fill it in.

5.2. Degree-equalized scalars. Call $f : \mathcal{M} \rightarrow \mathbb{R}$ *homogeneous of degree d* if $f(\lambda \cdot T) = \lambda^d f(T)$ for every $\lambda > 0$. Every scalar currently in the registry is homogeneous in this sense: angles have $d = 0$; side lengths, R, r , perimeter, and every point-to-point distance have $d = 1$; area has $d = 2$. Given a candidate triple f_1, f_2, f_3 , all everywhere-positive and of degrees $d_1, d_2, d_3 > 0$, define

$$\Phi = (f_1^{1/d_1}, f_2^{1/d_2}, f_3^{1/d_3}) : \mathcal{M} \rightarrow (\mathbb{R}_{>0})^3. \quad (5.2)$$

The point of the d_i -th root is that it equalizes scaling behavior: $f_i(\lambda \cdot T)^{1/d_i} = (\lambda^{d_i} f_i(T))^{1/d_i} = \lambda f_i(T)^{1/d_i}$, so *every* component of Φ scales by the same factor λ , regardless of the original d_i . Consequently $\Phi(\lambda \cdot T) = \lambda \Phi(T)$, so normalizing to unit length descends to a well-defined map on the quotient,

$$\bar{\Phi} : \Sigma \longrightarrow S^2_+, \quad \bar{\Phi}([T]) = \Phi(T) / \|\Phi(T)\|, \quad (5.3)$$

valued in the positive octant S^2_+ of the unit sphere. (Without the root-normalization step, the three components of Φ would scale at three different rates and no such map would descend to Σ ; this is the one place the homogeneity assumption is doing real work.)

5.3. The expected dichotomy. Both Σ and S^2_+ are 2-dimensional, so this is the $k = 3$ argument of Section 1 one dimension down, with $\mathcal{M}$ replaced by Σ and $\mathbb{R}^3$ replaced by S^2_+ : generically $D\bar{\Phi}$ has full rank 2 and, by the inverse function theorem, $\bar{\Phi}$ is a local diffeomorphism, so its image is open in S^2_+ . An exact homogeneous relation $G(f_1, f_2, f_3) = 0$ forces $\operatorname{rank} D\bar{\Phi} \leq 1$ everywhere, collapsing the image onto a curve — not generic, and detectable in the same spirit as Propositions 2.1–2.2, now with Σ and S^2_+ in place of $X \subset \mathbb{R}^3$, and geodesic distance on S^2_+ in place of Euclidean distance in $\mathbb{R}^3$.

Figure 5 shows this dichotomy directly: 4000 triangles sampled evenly over Σ , mapped through $\bar{\Phi}$ and viewed under a generic chart of the target (a stereographic projection, as in Remark 5.1, but on S^2_+ rather than Σ) — for a triple satisfying an exact relation, a curve; for one that doesn't, an open patch.

Remark 5.2 (Why this removes the null). Chordal (equivalently, up to a monotone reparametrization, geodesic) distance on S^2_+ is bounded above by the sphere's diameter, independently of which triple (f_1, f_2, f_3) is tested — unlike L_{real} in

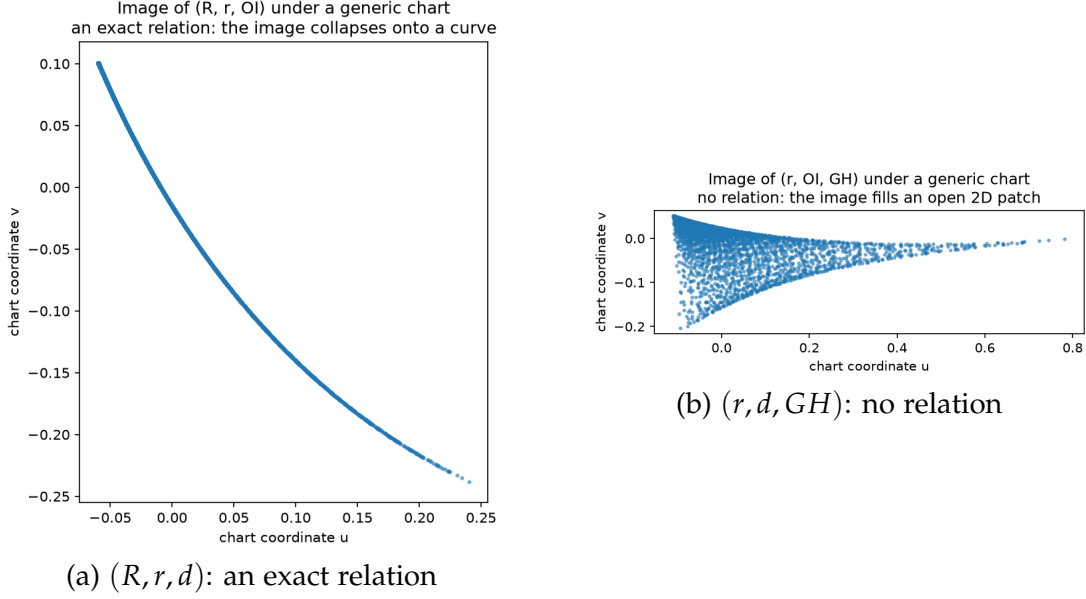


FIGURE 5. The image of $\bar{\Phi}$ under a generic chart of S_+^2 (Remark 5.1's construction, applied to the target). (a) The Euler triple (R, r, d) : an exact homogeneous relation collapses the image onto a 1-dimensional curve. (b) (r, d, GH) (inradius, incenter–circumcenter distance, centroid–orthocenter distance), with no known relation: the image fills an open 2-dimensional patch, as expected generically.

Section 2, whose scale depends on the raw units and typical magnitude of the f_i , which is precisely why Section 3 needed a triple-specific calibration. Realizing this in code, however, is not simply a matter of calling the bottleneck trainer of Section 2 with different data: that trainer (built on `scikit-learn`) always minimizes plain Euclidean squared error, with no hook for a different loss, and Euclidean distance in a flat chart of S_+^2 is *not* chordal distance on S_+^2 — a stereographic chart distorts distances, worse away from its excluded pole (Remark 5.1). `triangle_relations.discovery.sphere_autoencoder` therefore trains a small network (PyTorch, for its autograd) whose encoder and decoder operate on 2D chart coordinates of S_+^2 , but whose loss maps the decoder's chart-coordinate output back through the differentiable *inverse* chart before comparing it to the true point: the quantity gradient descent actually minimizes is chordal distance on S_+^2 itself, $\|\bar{\Phi}(T) - \psi(d(e(\varphi(\bar{\Phi}(T))))\|$ where φ, ψ are the chart and its inverse, not a chart-space proxy for it. Squared chordal distance is bounded in $[0, 4]$ for *every* triple with no rescaling needed, since $\bar{\Phi}(T)$ is already on a fixed, triple-independent scale by construction (a unit vector) — so, with no null and no dependence on a shuffle count m . This also allows Σ to be sampled evenly and deterministically (Remark 5.1), rather than relying on i.i.d. random triangles to cover it.

Remark 5.3 (Why a generic chart is safe). A chart is only ever defined on the complement of some lower-dimensional excluded locus — a single point for the ζ -chart of Remark 5.1; more generally, for a chart on $\mathbb{RP}^2$ itself, a full line "at infinity." This raises an obvious worry: what if the image we are trying to observe lands entirely inside the excluded locus of whichever chart we happened to pick, invisible to it? For

a *generic* choice of chart this cannot happen when there is no relation: the excluded locus has positive codimension, while the image of a relation-free triple is open (the same dichotomy as Section 1), so it can only meet a fixed positive-codimension subset in a measure-zero way, never entirely. And if the image *were* confined to the excluded locus regardless, that would not be a failure of the method — it would mean the degree-equalized coordinates satisfy a further, linear relation (confinement to a proper linear subspace), exactly the kind of coincidence this apparatus is built to notice, just a simpler one. On the target side, the current implementation picks a single fixed pole for every triple — outside the positive octant every positive-degree scalar triple’s image lives in (every registered scalar of positive degree is strictly positive on every nondegenerate triangle), so its excluded locus never comes close to any actual data, for any triple (`spherical_chart.CANONICAL_POLE`). Since the pole is fixed rather than chosen per triple, the chart itself introduces no per-triple calibration, consistent with Remark 5.2.

Remark 5.4 (Scope: degree-0 scalars). This construction does not cover degree-0 scalars (angles), since $f^{1/0}$ is undefined; `search_homogeneous_relations` excludes them from the search (logged at run time), leaving them to Program 1’s more general, but more expensive, permutation-null test.

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- [1] D. G. Kendall, *Shape manifolds, Procrustean metrics, and complex projective spaces*, Bull. London Math. Soc. **16** (1984), 81–121.